

5. Vanishing Gradient Problem

[DL](#)

Vanishing Gradient Problem occurs when gradients become extremely small during backpropagation in deep neural networks, hindering training of early layers. This issue arises primarily from repeated multiplications of small derivatives in activation functions like sigmoid or tanh as signals propagate backward

Why Does It Happen?

1. Repeated Multiplication of Small Numbers
2. **Saturating Activation Functions**
Activation functions like:
 - **Sigmoid**
 - **Tanh**have derivatives close to **0** for large positive or negative inputs.

How to Solve / Reduce It

1. **Use Better Activation Functions**
 - ReLU
 - Leaky ReLU
 - ELU
2. **Proper Weight Initialization**
 - **Xavier (Glorot) Initialization** → for tanh/sigmoid
 - **He Initialization** → for **ReLU**
3. **Batch Normalization** -> Normalizes layer inputs
4. Reduce Model Complexity

[**Code**](#)